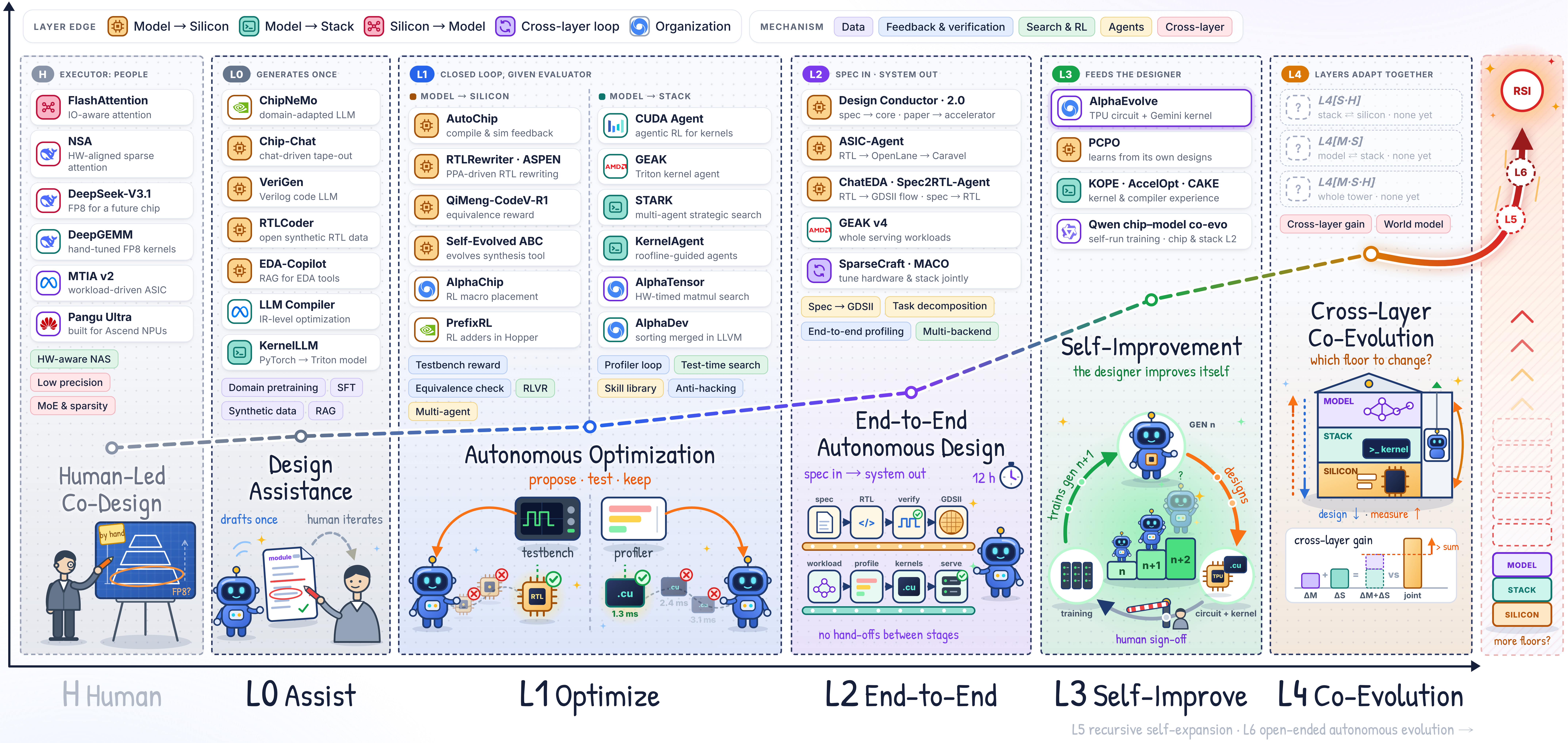

LAYER EDGE: Model → Silicon · Model → Stack · Silicon → Model · Cross-layer loop · Organization

MECHANISM: Data · Feedback & verification · Search & RL · Agents · Cross-layer

H EXECUTOR: PEOPLE

- **FlashAttention** IO-aware attention
- **NSA** HW-aligned sparse attention
- **DeepSeek-V3.1** FP8 for a future chip
- **DeepGEMM** hand-tuned FP8 kernels
- **MTIA v2** workload-driven ASIC
- **Pangu Ultra** built for Ascend NPUs

HW-aware NAS · Low precision · MoE & sparsity

Human-Led Co-Design

L0 GENERATES ONCE

- **ChipNeMo** domain-adapted LLM
- **Chip-Chat** chat-driven tape-out
- **VeriGen** Verilog code LLM
- **RTLCoder** open synthetic RTL data
- **EDA-Copilot** RAG for EDA tools
- **LLM Compiler** IR-level optimization
- **KernelLLM** PyTorch → Triton model

Domain pretraining · SFT · Synthetic data · RAG

Design Assistance

drafts once · human iterates

L1 CLOSED LOOP, GIVEN EVALUATOR

MODEL → SILICON

- **AutoChip** compile & sim feedback
- **RTLRewriter · ASPEN** PPA-driven RTL rewriting
- **QiMeng-CodeV-R1** equivalence reward
- **Self-Evolved ABC** evolves synthesis tool
- **AlphaChip** RL macro placement
- **PrefixRL** RL adders in Hopper

Testbench reward · Equivalence check · RLVR · Multi-agent

MODEL → STACK

- **CUDA Agent** agentic RL for kernels
- **GEAK** Triton kernel agent
- **STARK** multi-agent strategic search
- **KernelAgent** roofline-guided agents
- **AlphaTensor** HW-timed matmul search
- **AlphaDev** sorting merged in LLVM

Profiler loop · Test-time search · Skill library · Anti-hacking

Autonomous Optimization

propose · test · keep

L2 SPEC IN · SYSTEM OUT

- **Design Conductor · 2.0** spec → core · paper → accelerator
- **ASIC-Agent** RTL → OpenLane → Caravel
- **ChatEDA · Spec2RTL-Agent** RTL → GDSII flow · spec → RTL
- **GEAK v4** whole serving workloads
- **SparseCraft · MACO** tune hardware & stack jointly

Spec → GDSII · Task decomposition · End-to-end profiling · Multi-backend

End-to-End Autonomous Design

spec in → system out · 12 h

no hand-offs between stages

L3 FEEDS THE DESIGNER

- **AlphaEvolve** TPU circuit + Gemini kernel
- **PCPO** learns from its own designs
- **KOPE · AccelOpt · CAKE** kernel & compiler experience
- **Qwen chip–model co-evo** self-run training · chip & stack L2

Self-Improvement

the designer improves itself

L4 LAYERS ADAPT TOGETHER

- *L4[S·H]* stack ⇄ silicon · none yet
- *L4[M·S]* model ⇄ stack · none yet
- *L4[M·S·H]* whole tower · none yet

Cross-layer gain · World model

Cross-Layer Co-Evolution

which floor to change?

design ↓ · measure ↑

more floors?

H Human · L0 Assist · L1 Optimize · L2 End-to-End · L3 Self-Improve · L4 Co-Evolution

L5 recursive self-expansion · L6 open-ended autonomous evolution →